

The One Introduction Problem

A supply-constrained sequential matching policy: clarification targeting, outcome-relevant weighting, and variance-aware evaluation

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Team: Bipartite Bard **Team leader:** Anadi Tripathi **Members:** Anadi Tripathi **Institution:** IIT Madras **Repository:** <https://github.com/Yash-Tripath1/sequential-matching-public-simulator-lab>

Declaration. We confirm that this submission represents our team’s work and that all external papers, datasets, code, AI tools and other resources used are acknowledged in §8.3 (AI tools and resources) and §8.5 (references). AI tools used: Arena.ai Agent Mode and additional large-language-model chat assistants (ideation, code debugging, drafting and pre-submission review); see §8.3.

Scope statement. Every person, preference, conversation and outcome in this work is synthetic content from the organisers’ public development kit. Nothing here estimates relationship compatibility for real people, and no result below is evidence of product effectiveness. All experiments run against the public simulator on public seeds; no private world, private seed or private organiser file was used. Analysis marked *analysis-only* reads simulator ground truth and is never imported by, or available to, any policy at inference time.

1. Abstract and thesis

Our own three-seed pilot reported that a “potential-ask” clarification heuristic raised MSMI per 100 arrived members from 0.389 to 0.639, and a ten-seed screen put the incumbent at 0.525 against 0.350 for the starter greedy baseline. We reproduce the incumbent’s 0.525 exactly on the same ten public seeds. Then we ask a question neither run could answer: **how much room is actually left?** We answer it by measuring the *structure* of the generated worlds rather than only the scores they produce. Under complete disclosure, the reciprocal hard-constraint graph of a 200-member development pool has mean degree **2.53**, feasible-pair density **1.27%**, and **33.9%** of members with no feasible partner at all. Because roughly a third of members also carry a *declined* hard field — which the contract says must never be imputed or bypassed — the fraction who could **ever** be introduced, under any policy with perfect clarification, is **0.399** in development, **0.341** in cold start and **0.178** in sparse geography. The number of distinct feasible pairs whose endpoints are never decline-blocked — a hard cap on total introductions, since pairs cannot repeat within an episode — is **118.0**, **90.6** and **23.3** respectively.

That reframes the problem. **The primary score in this simulator is supply-determined, not ranking-determined.** Any asking policy that reaches the coverage ceiling has already captured the largest available lever, and the residual differences between matchers operate on a matching that is close to forced: with median degree 2 there is little choice to optimise. Our own earlier ten-seed screen pointed the same way without being able to say why — it estimated a clarification main effect of +0.117 MSMI/100 against a matcher main effect of +0.058 — and the structural measurement explains the asymmetry.

We therefore argue that the productive question is not “which policy ranks pairs best?” but “**where does the attainable score actually come from?**” Our answer, from a three-way factorial of ask rules × pair-weight rules × allocation rules at ten seeds per family, plus an oracle ladder, is that it comes from three places in strict order of size — and that the place most teams invest in is the smallest.

The strongest single number we can defend is not our best cell. It is the **pre-specified** comparison of our existing incumbent against the starter baseline, fixed by the earlier screen before the new factorial ran: `lab_potential_maxw` scores **0.525 versus 0.350** for the official greedy baseline, a paired difference of **+0.175 MSMI/100**, **95% seed-block CI [+0.092, +0.258]**, **p = 0.016**, **23 wins / 29 ties / 8 losses** over ten seeds × six families. Everything else in §6 is screening, and is labelled as such.

Caveat: the ten evaluation seeds are the same ten used by the earlier screen that selected this incumbent, and seeds 101/202/303 were also the three-seed pilot on which it was first chosen. This comparison is therefore a recomputation on selection data, not an independent confirmation.

Thesis. With reciprocal hard feasibility enforced, MSMI is determined first by *whether a policy makes the feasible graph usable at all* — a step function that any clarification rule, however naive, saturates; second by *how much of the resulting edge supply it consumes* before members exit or retire — which we measure at 68–87% and which no rule we tested moved; and only third, within a band narrower than 60 episodes can resolve, by *how well it*

ranks the pairs it gets to choose between. Targeting clarification at expected unlock value, fitting a calibrated pair-outcome model, adapting online on dense feedback and spending the residual budget on soft answers are all third-order interventions, and we report that none of them beat the incumbent — one of them, a correctly calibrated probability used as an edge weight, scores below a hand-coded similarity score in all three ask cells (suggestive only: 3 of the 78 pairwise intervals exclude zero, about what chance alone gives), with a candidate explanation in §6.2c. The two changes that did move the score are not models: correcting the *coding* of the pair weight so that a known disagreement counts as negative and every feasible edge keeps a positive weight, and replacing greedy ordering with global matching. Each is worth about half of their combined +0.133 MSMI/100, and neither is individually significant at ten seeds.

Stated as falsifiable hypotheses and resolved against our own runs:

#	Claim	Verdict
H1	Unlock-value ask targeting beats count-based and unranked asking at a fixed matcher	Not supported — differences −0.000 to +0.008 at a fixed matcher, all CIs straddling zero
H2	Any asking policy beats no clarification	Supported — +0.183 MSMI/100, CI [+0.042, +0.325]
H3	Online adaptation on dense directional responses helps in drift/shift	Falsified as implemented — 0 wins / 60 ties / 0 losses, MSMI outcomes identical to its static version
H4	Waiting/batching helps in supply-rich families and not in sparse	Untested — planned (§8.1)
H5	Residual soft-field clarification does <i>not</i> help, because the matching is near-forced	Not rejected — +0.017, CI [−0.083, +0.158], $p = 0.961$ (interval too wide to establish equivalence)
H6	Coverage of any valid policy converges to the structural ceiling	Supported — every asking policy at 97–98% of it, and an oracle with free perfect clarification reaches only 0.1 pp more

The most consequential of these is H6, because it is confirmed from two directions. From below: all sixteen asking configurations make 4,555–4,681 introductions over their 60 episodes and land at coverage 0.340–0.343 — a spread of 0.003, about 1% relative, across rules that differ in every design choice we could think of. From above: an *oracle* given free, complete, instantaneous disclosure of every non-declined field — hard and soft — reaches coverage **0.298** against **0.297** for a random-feasible policy on the same thirty dev/sparse/cold worlds, and **62.9** assignments against **61.9**; pushing further to a maximum-*cardinality* matching under perfect information moves those to 0.299 and 63.1. **Perfect information buys a tenth of a percentage point of coverage and one introduction per episode**, and it buys *fewer* assignments than the non-oracle configuration at rung R3 of the same ladder manages on the same worlds (63.2) — a caveat on that comparison is in §6.6. What it does buy is a better-selected set — the mutual-acceptance rate per assignment rises from 0.144 to 0.161 — but the primary score still *falls*, from 0.300 to 0.233, because the extra selectivity is paid for with introductions that never happen. Whatever gap remains between a submitted policy and the attainable bound is not an information gap, and a clarification strategy is not where it will be closed.

Three secondary results carry the mechanism, and are developed in §4.3 and §6: **four of the seven soft fields carry no detectable outcome signal** (agreement on relationship_goal doubles mutual acceptance, 0.150 → 0.312, while lifestyle, emotional_availability, space_for_relationship and relocate sit at ratios of 0.92–1.16); **observability is a confound, not a quality signal** — mutual acceptance is *lower* when more soft fields are known, 0.2724 with none versus 0.2317 with all seven, which is why a restricted four-predictor prior beats a fuller eighteen-feature one on every held-out measure; and **the six families are not drawn from one outcome function** while the policy cannot observe which family it is in, so a single offline weight vector is misspecified for at least two of them. We also report what our evidence does *not* support. The potential-ask advantage survives at ten seeds against the starter greedy baseline but its interval against the stronger default-ask control at a fixed matcher includes zero, its sign-flip test is unadjusted for the dozen contrasts we ran, and it **reverses in cold start**. Our fitted outcome prior does not beat a hand-coded signed similarity score. Our online adaptive learner produced MSMI outcomes identical to its own static version in all 60 episodes (0/60/0), while its assignments differed slightly (4,658 vs 4,652 in total) — so its *decisions* were not bit-identical, only its outcomes were. The phase hybrids screened earlier are exploratory and subject to winner’s curse. And a zero MSMI count in a low-supply family is not evidence that two policies are equivalent.

Contributions. (i) A structural characterisation of the generated worlds — coverage ceilings, edge-supply caps and the binding-constraint histogram — that turns “why is my score low?” into a measurable quantity. (ii) An expected-unlock-value clarification rule with an empirical pass-proability prior estimated from the supplied training pools only. (iii) A restricted, calibration- checked pair-outcome

prior fitted on self-logged randomised rollouts, with an explicit treatment of missing answers as unknown rather than as disagreement. (iv) A residual-budget soft-field clarification rule that no supplied baseline uses. (v) A variance-aware evaluation protocol with seed-block bootstraps, exact sign-flip tests and a power analysis showing what the official 120-episode assessment can and cannot resolve.

2. Formalisation

2.1 Notation

An episode runs over decision days $t = 0 \dots 59$ followed by 40 follow-up days with no new introductions. Let V be the members of one pool, $|V| = 200$. Member i has an arrival day $a_i \leq 20$, an exit day e_i , an always-observable triple $(\text{age}_i, \text{gender}_i, \text{zone}_i)$, and a questionnaire $\{f_{ik}\}_{k \in H \cup S}$ where H is the eleven-field hard bundle and S the seven soft fields. Each field carries a value, a status in $\{\text{observed}, \text{not_asked}, \text{declined}\}$, and an observation day. Write $f_{ik} = \emptyset$ when the value is unknown. $N = |\{i : a_i \leq 59\}|$ is the scoring denominator and includes members who later exit or are never served. The primary metric is $\text{MSMI}/100 = 100M/N$ where M counts qualifying introductions, one per pair. The official score is the unweighted mean of six family means, each over twenty private seeds.

2.2 Three-valued reciprocal feasibility

The supplied eligibility(a , b) returns one of three values, and the distinction is the hinge of the whole problem:

- **infeasible** — at least one *observed* hard check fails in either direction. No amount of clarification can rescue this pair.
- **needs_clarification** — no observed check fails, but at least one hard field is $\emptyset$ for one of the two members. The pair is *blocked*: it may be feasible or infeasible, and the policy cannot know which.
- **feasible** — every hard field is observed for both members and every reciprocal check passes. Only these pairs may be proposed.

Reciprocity means each check is evaluated twice, once with each member as the constraint holder: j 's age must lie in i 's stated window **and** i 's in j 's; j 's zone must be in i 's acceptable_zones **and** vice versa; partner_smoking and partner_children are one-directional preferences applied in both directions; relationship_structure, wants_children and schedule are symmetric. A favourable score cannot override any of these.

Two consequences deserve emphasis because they are easy to get wrong.

Declined is absorbing. resolve_asks will not overwrite a declined field, and a $\emptyset$ hard field forces needs_clarification. A member with even one declined hard field can therefore never appear in a feasible pair, on any day, under any policy. In the ten public development worlds, **31.9%** of members are in this state; in cold start, where fewer fields are pre-observed and so more chances to be declined arise, **41.0%**. Spending clarification units on such a member is a guaranteed waste, and every supplied baseline's ask filter already excludes them — but no published analysis quantified how much of the pool that removes.

Zones are opaque labels with no adjacency. The check is set membership only. relocate is a soft field and does not override acceptable_zones.

2.3 The outcome funnel

An introduction produces a chain of five events, and the spec is explicit that they must not be collapsed:

assigned $\rightarrow$ both respond $\rightarrow$ mutual accept $\rightarrow$ date $\rightarrow$ both second-Yes, on time, in window

MSMI requires a date within 30 days of assignment **and** both members returning Yes within 3 days of that date. Our measured stage rates on 7,613 self-logged introductions are:

Stage	Estimate	Numerator / denominator	Controllable by the policy?
Both members respond by the deadline	0.5932	4,516 / 7,613	No (member-level response propensities)
Mutual acceptance, given both responded	0.2573	1,162 / 4,516	Yes — via pair selection
Date happens, given mutual acceptance	0.7685	893 / 1,162	No
MSMI, given a date	0.0907	81 / 893	Partly — relationship_goal agreement enters this stage again
MSMI, given an assignment	0.0106	81 / 7,613	—

Two of the four transitions are outside the policy's control, and they are the large losses: 41% of

introductions never get two responses, and 91% of dates fail the on-time double-Yes (only 204 of 893 dates produce two answers inside the 3-day window). The 3-day window against a 1–5 day answer delay alone caps that last transition at $0.6^2 = 0.36$. **The realistic ceiling on MSMI per assignment is therefore of order 10^{-2}** , which is why a 200-member episode yields about one event and why single-episode comparisons are meaningless.

2.4 MSMI as a rare, delayed, self-censoring event

Three properties drive the methodology:

- **Rare.** One event moves a 200-arrival episode's score by 0.5. Pooled across all our episodes the per-episode SD is reported in §6.4.
- **Delayed.** Directional responses are visible within 7 days; a date within roughly 21 (up to 33 in delayed); the second-meeting answers a few days after that. An MSMI label typically matures 30+ days after the assignment that earned it, inside a 60-day decision horizon, with policy memory reset at every episode boundary.
- **Self-censoring.** Feedback exists only for pairs the policy chose. The supplied propensity field is null, so no unbiased inverse-propensity claim can be made about the supplied files. We address this by generating our own randomised logs (§4.3), not by re-weighting theirs.

There is one further dynamic that a static analysis misses: when both members return a second-meeting Yes, the simulator records `pause_after_mutual_interest` and **both leave the market permanently**. Success consumes supply. Since supply is the binding constraint (§3), the policy is not simply maximising a per-day objective — every introduction spends a scarce, non-renewable resource.

2.5 What the policy may observe

Only `observe()` output: arrived members with values, statuses and observation days; the introduction list; feedback whose `observed_day` has been reached; the cumulative ask log; the day; remaining budget. No world seed, no latent preference, no response propensity, no arrival forecast, **no scenario label** — which is why §4.5's non-stationarity problem has to be solved online rather than by looking up the family. We use exactly this, plus offline-fitted assets derived from our own rollouts on declared training seeds.

3. Structural diagnosis (analysis-only)

This section reads simulator ground truth in order to measure the feasible graph. It is a measurement instrument. No policy imports it, and the numbers it produces are never available to a policy at inference time.

3.1 The feasible graph is thin

Ten public seeds per family, complete disclosure, members arrived by day 59:

Family	Feasible-pair density	Mean degree	Median degree	Members with degree 0	Decline-blocked	Coverage ceiling	Effective edges	Max same-day matching
development	0.0127	2.53	2	33.9%	31.9%	0.399	118.0	34.4
sparse	0.00245	0.49	0	67.0%	32.8%	0.178	23.3	15.3
cold start	0.01337	2.66	1	32.1%	41.0%	0.341	90.6	28.6
delayed / shift / drift	identical to development					0.399	118.0	34.4

Coverage ceiling = the fraction of arrived members with at least one feasible partner who is also never decline-blocked. It is the value a policy with free, perfect, instantaneous clarification would reach. *Effective edges* = feasible pairs with neither endpoint decline-blocked; because a pair cannot repeat within an episode, this is an upper bound on total introductions.

The sparsity of sparse is now quantitative rather than anecdotal. Twelve opaque zones instead of four, with acceptable_zones never expanded beyond a member's own zone, drives **67.0%** of members to degree zero and leaves **23.3** usable pairs in the entire pool. Geography displaces who_to_meet as the top binding constraint (18,176 of 19,900 pairs in seed 101, against 10,974 in development). With ~20 assignments per episode at the measured funnel rates, the expected MSMI in a sparse episode is about 0.24 events, i.e. **0.12 per 100** — so the published zeros and near-zeros are what a supply-limited family produces, not a policy failure.

3.2 Which constraint actually binds

Development, seed 101, all 19,900 pairs under complete disclosure:

Blocking reason	Pairs blocked	Share of 19,900
who_to_meet	14,757	74.2%
age window	12,116	60.9%
geography	10,974	55.1%
wants_children conflict	4,761	23.9%
smoking	4,545	22.8%
relationship_structure	3,600	18.1%
partner_children	2,970	14.9%
schedule	2,684	13.5%
feasible	273	1.37%

Reasons overlap, so shares do not sum to one. The ordering matters for clarification targeting: who_to_meet, age and geography are the three gates that eliminate most pairs, and all three are cheap to reason about because the relevant quantities — a member's own age, gender and zone, and the partner's — are **always observable**. Only the constraint holder's answer is unknown.

3.3 Empirical pass-probability prior

Estimated from the six supplied **training** pools only (public_01-public_06, 1,200 members, per data_manifest.json), using only disclosed answers:

Unknown field	Pass probability	Notes
who_to_meet	0.494 / 0.480 / 0.461	by partner gender: woman / man / non-binary
acceptable_zones, same zone	1.000	own zone is always accepted
acceptable_zones, different zone	0.0756	only the 22.9% of members with two zones can pass
schedule overlap	0.856	
relationship_structure equal	0.793	88.3% monogamous
wants_children no conflict	0.755	yes 0.362 / no 0.339 / unsure 0.300
smoking acceptable	0.878	50.4% accept any; 75.4% are non-smokers
partner_children acceptable	0.921	15.5% have children
age window	~0.13	marginal used by the unlock prior; see note below

Age-window conditioning. Age itself is always observable, so the age window is never an *unknown* field in the eligibility check, and §3.2's figure (age blocks 60.9% of all fully-disclosed pairs, a ~0.39 pass rate over all 19,900 pairs) is a different quantity: it is measured on the full pair population. The ~0.13 row is the marginal the unlock-value rule uses when scoring blocked *candidate* pairs under the §4.2 prefilter (same zone, |age gap| ≤ 10), estimated as described in scripts/estimate_priors.py from the six training pools. The two numbers answer different questions and must not be compared directly.

The same-zone / different-zone asymmetry is the sharpest fact in the table: a blocked cross-zone pair carries roughly $0.076^2 \approx 0.006$ of the unlock probability of a blocked same-zone pair, before any other constraint is considered. Any clarification rule that ranks members without conditioning on zone wastes most of its budget.

4. Method

We keep normalisation, modelling and allocation separate, as the integration requirement asks. Component boundaries are explicit so each can be ablated.

4.1 Allocation: maximum-weight matching on a general graph

Why this is a matching problem at all. A member may hold **at most one concurrent, outstanding introduction** — not one per day, and not one for the whole episode. Two members who are both already under an outstanding introduction cannot be assigned to anyone else until it resolves. That single rule is what makes the per-day decision a *matching* on the feasible graph rather than an inde-

pendent ranking of pairs: choosing edge (i, j) forecloses every other edge incident on i or on j for as long as the introduction is outstanding. Any allocation rule that scores pairs independently and takes the top- k will therefore propose collisions that the simulator must reject, and the rejected slot is wasted supply.

The feasible graph is **not bipartite** — gender preferences are stated per member and need not partition the pool — so assignment/Hungarian methods do not apply. We use `networkx.max_weight_matching`, which implements Edmonds’ blossom algorithm (§8.5) for general graphs.

The per-day objective is $\max_{\mathcal{M}} \sum_{(i,j) \in \mathcal{M}} \hat{p}_{ij}$ where $\hat{p}_{ij}$ estimates $\Pr(\text{MSMI} \mid i, j \text{ assigned})$ and $\mathcal{M}$ is a matching. **We state the independence assumption explicitly, as the brief requires.** Pair outcomes are *not* exactly independent: the generator adds a per-pair shared term, and we measure the ratio of observed joint acceptance to the product of the two directional marginals at **1.105**, so directional probabilities must not be multiplied without that caveat. Given the world, however, the residual dependence is weak enough that expected MSMI is *approximately* linear in the chosen edges, which is what makes maximum-weight matching on $\hat{p}$ a good single-day objective. We report $\Pr(\text{mutual accept} \mid \text{both responded within 7 days})$ as our estimation target throughout, and never conflate it with $\Pr(\text{MSMI})$ or with either directional probability; calibration of that target on held-out seeds is in §4.3.

One honest qualification, which §6.2c then tests and **refutes for our own fitted model**: if $\hat{p}$ were the true outcome probability, maximum-weight matching on it would be the exact single-day optimum. Two implementation points follow, and both were bugs we had to fix:

- **Weights must be positive.** A signed similarity score that goes negative causes the matcher to *drop* the edge, forfeiting an introduction whose true $\Pr(\text{MSMI})$ is small but strictly positive. Under a supply constraint that is a pure loss. Our weights are probabilities, or logistic transforms of similarity, and are never ≤ 0 .
- **Unknown is not disagreement.** Coding a known mismatch and an unknown answer both as 0 — which an “agreement count” scorer does — makes the matcher indifferent between “we checked and they clash” and “we have no idea”. Our coding is $+1 / -1 / 0$, and the fitted intercept absorbs the unknown case, which is the correct expected value under the prior.

We do not claim a sequential optimum. Waiting has value when a better partner may arrive and cost when a member exits or a rival consumes the pair; with median degree 2 the choice set is small, so we treat waiting as a separate hypothesis (H4, §5.2) rather than folding it into the matcher.

4.2 Clarification: expected unlock value

For each askable member i we score

$$V_i = \frac{1}{3} \sum_{j \in \mathcal{C}(i)} \pi_{ij} \cdot \max(0, w_{ij} - u_j)$$

and propose the highest- V members while the daily budget allows.

- $\mathcal{C}(i)$ — candidate partners: same zone, $|\text{age}_i - \text{age}_j| \leq 10$, and no *known* hard failure. Members with any unknown hard field have no feasible edge, so they are absent from the current matching and their own dual is zero; only j ’s opportunity cost survives.
- π_{ij} — probability that every still-unknown hard check passes, computed as a product of the §3.3 marginals over exactly the fields that are unknown for this pair. This is an empirical prior from disclosed answers, not a generator constant.
- w_{ij} — the pair’s matching value on observable soft fields (§4.3).
- u_j — j ’s opportunity cost: the weight of j ’s edge in yesterday’s maximum-weight matching, or 0 if j was unmatched. The term $\max(0, w_{ij} - u_j)$ is a **reduced profit**: unlocking an edge that would not displace anything is worth nothing.

This is the shadow-price idea from our earlier plan, with two honest simplifications. True duals on a general graph involve blossom variables that `networkx` does not expose, so u_j is an approximation from the realised matching rather than an exact vertex dual. And the same-zone / age-gap prefilters discard cross-zone candidates whose unlock probability is $\sim 10^{-3}$ of a same-zone one; this is an approximation made for the 10-second invocation limit, and we report it as such.

Never re-ask, never impute. Members with a declined hard field are excluded from the candidate set entirely; declined soft fields are excluded from soft asks. A null is unknown, never zero, never a rejection.

4.3 Modelling: a restricted pair-outcome prior

Training data. We ran a *randomised logging policy* on eighteen declared training seeds (11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79), disjoint from the ten evaluation seeds: unlock-value clarification plus uniformly shuffled feasible matching. Randomised matching is what

makes the log usable — pair quality spans its whole observable range instead of concentrating on one matcher’s preferences. This produced 108 episodes and **7,613 labelled introductions**, of which 4,516 had both members respond. Features are recorded at assignment time; labels are harvested only after the 40-day follow-up, so no post-action observation is attached to an earlier decision.

Target. $\Pr(\text{mutual acceptance} \mid \text{both responded})$ over the 7-day response window — the densest outcome-relevant stage. We fit MSMI directly as well, and report why we do not use it as the ranking target: with a 1.13% base rate the direct model achieves held-out AUC 0.581 and log loss 0.0611 against a null of 0.0618, i.e. essentially no discrimination. The mutual-acceptance stage carries the same pair-level signal at 40× the label density, and the two stages that follow it are close to pair-independent apart from a further `relationship_goal` effect that the acceptance model already captures.

Specification search. Three nested specifications, each fitted by ridge logistic regression (IRLS, $\lambda = 3$, intercept unpenalised) and evaluated on six held-out training seeds:

Specification	Features	Held-out log loss	Null	Brier	AUC
A	7 agreement indicators	0.55198	0.56650	0.18337	0.595
B (shipped)	4 core agreement indicators	0.55125	0.56650	0.18298	0.598
C	A + 7 both_known + age gap + same-zone + day	0.55722	0.56650	0.18561	0.574

Specification C loses on every held-out measure despite having four times the parameters. The reason is diagnostic and worth stating plainly, because it is a trap this dataset sets for any model that encodes missingness as a feature:

- Mutual acceptance is **0.2724** when no soft field is known for the pair, 0.2546 when one to three are known, and **0.2317** when all seven are known. Observability is negatively associated with acceptance, because whether a member’s answers are pre-observed is an independent draw in the generator and carries no preference information.
- The seven `both_known` indicators are near-collinear (a member either has a complete profile or does not), so the fit assigns them large offsetting coefficients — one reached +0.61 — that encode *who is observable*, not *who is compatible*.
- `same_zone` has 4,492 observations on one side and **24** on the other among logged pairs; its coefficient is unidentifiable and absorbs intercept shift.

The shipped prior is therefore deliberately small:

Coefficient	Value
intercept	−1.0065
<code>relationship_pace</code> : agree	+0.5328
<code>relationship_goal</code> : agree	+0.4212
<code>conversations</code> : agree	+0.2958
<code>lifestyle</code> : agree	−0.0702

Calibration on held-out data. The brief asks that any probability estimate define its target and observation window and then be shown calibrated out of sample. Target: mutual acceptance given both members responded within the 7-day window. Fitted on 12 seeds (3,054 both-responded pairs), evaluated on the six disjoint held-out seeds (1,462 pairs, base rate 0.2538). The table below reports the *extended* specification C, whose wide predicted range exposes the missingness confound; the shipped four-feature specification B has held-out log loss 0.5513 and Brier 0.1830 on the same split:

Predicted bin	<i>n</i>	Mean predicted	Observed rate	Gap
0.05–0.10	70	0.0897	0.1571	+0.067
0.10–0.20	248	0.1694	0.1532	−0.016
0.20–0.30	784	0.2631	0.2640	+0.001
0.30–0.40	308	0.3319	0.2922	−0.040
0.40–0.60	51	0.4736	0.4706	−0.003
0.60–0.80	1	0.6003	1.0000 (n = 1, uninformative)	

The model is well calibrated across the 1,461 pairs that matter — the largest absolute gap in a bin with $n \geq 50$ other than the lowest is 0.040, and the three central bins holding 1,340 pairs are within 0.040 of nominal. The one real defect is the lowest bin, where 70 pairs are predicted at 0.090 and come in at 0.157: the ridge prior over-shrinks strongly disagreeing pairs toward zero. Held-out log loss 0.5572, Brier 0.1856 for the extended specification shown (spec B: 0.5513 / 0.1830). This is also why the note is careful about the direction of the \$6.2c result — the fitted prior is calibrated, and calibration is not what makes a good edge weight here.

The coefficients are used only to rank and weight feasible edges. Probability estimates are optional under the competition rules and do not affect ranking.

Directional probabilities are not multiplied. “A accepts B”, “B accepts A” and “both accept” are different quantities and the two directions are positively dependent: on held-out pairs the observed mutual-acceptance rate is **0.2538** while the product of the fitted directional probabilities is **0.2297**, a ratio of **1.105**. We therefore model the joint directly. Any submission that reports directional probabilities must state this dependence rather than assume independence.

4.4 Residual-budget soft clarification

This component is new relative to every supplied baseline and to our own pilot. Two measurements motivate it:

- The 12-unit daily budget is **not** the binding constraint. Measured ask spend is ≈ 195 units per episode against a 720-unit ceiling, and every asking method we tested lands within a few units of that same figure. It is not under-spending: 31.9% of development members are decline-blocked and so never askable, 36.4% already have every hard field observed at arrival, and the remaining askable population is exhausted well before the horizon (\$6.8). The *askable population*, not the budget, runs out.
- Outcome-relevant soft fields are observed for only $\sim 46\%$ of members in development and $\sim 34\%$ in cold start. No baseline spends a single unit on them.

Once hard-bundle asks are exhausted, the rule converts residual units into 1-unit soft answers, prioritising (member, field) pairs by $\deg(i) \times (0.2 + |\beta_k|)$ — partner availability times the field’s fitted influence. In development this lifts realised ask spend from 195 to ~ 558 units per episode. Whether it converts into MSMI is an empirical question with a real prior against it: at median degree 2 the matching is nearly forced, so better weights may have little room to act. We report the measured effect in §6 rather than assuming it.

4.5 Online adaptation on dense proxies

Every episode starts with empty memory, and MSMI matures 30+ days after assignment — about one positive label per episode. A bandit that updates only on MSMI is structurally starved, which is our explanation for the earlier Thompson and GP results rather than a claim that bandits are inappropriate.

Directional responses are the alternative: two per introduction, visible within seven days, roughly 120 usable observations by mid-episode. We refit the §4.3 model online by ridge-anchored IRLS towards the offline prior ($\lambda = 4$ toward β_0 , minimum 40 observations, most recent 1,200 retained), so a thin in-episode sample cannot drag the model away from its training. Non-responses are **excluded** from the update, never recoded as rejections.

For the drift and shift families we additionally admit a single day/60 coefficient that is *zero in the offline prior* and estimated only in-episode. The offline prior pools stationary and drifting families, so a day term is not identifiable there; within one episode the family is fixed, so it is. Evidence that there is something to detect: mutual acceptance in drift falls $0.263 \rightarrow 0.237 \rightarrow 0.164$ across day blocks (<30 , $30\text{--}44$, ≥ 45) while development is flat at $0.263 \rightarrow 0.273 \rightarrow 0.246$.

Carried memory stays well inside the 1 MiB limit: a coefficient vector, a bounded feature buffer and counters. We will verify the serialised size before Round 2 rather than assume it.

4.6 Audit ledger

For every proposed pair we emit the observed evidence per field (value, status, observation day), unresolved fields, the binding constraint when a pair was rejected, the runner-up pair and its weight, and the reduced profit $w_{ij} - u_i - u_j$. Explanations separate three things that must not be conflated: **model uncertainty**, **missing data** (a field that is $\emptyset$ or declined), and **stochastic outcome**. A latent simulator value is never presented as an observation and no personal history is invented.

4.7 What we deliberately did not do

- **No hidden-state access at inference.** No policy reads `world['truth']`, reconstructs answers from generator seeds or IDs, or inspects organiser files.
- **No latent-probability oracle rung.** Our earlier plan proposed an “expected-outcome oracle”. The simulator does not expose latent pair probabilities, so that rung would require reconstructing the generator’s outcome function. We dropped it. The oracle ladder in §6.6 stops at perfect clarification plus maximum-cardinality allocation.
- **No inverse-propensity claims on the supplied files.** The propensity field is null. Doubly-robust off-policy evaluation on our own randomised logs is planned work (§8.1), not a result.
- **No Thompson-sampling-plus-value-of-information hybrid.** Not implemented; it would be new work, and we do not describe it as though it exists.

5. Experimental design

5.1 Worlds, seeds and splits

Use	Seeds	Families	Episodes per method
Prior fitting / model selection	11, 13, 17, 19, 23, 29, 31, 37 (fit) and 41, 43, 47, 53, 59, 61, 67, 71, 73, 79 (holdout)	all six	108 logged
Head-to-head evaluation	101, 202, 303, 404, 505, 606, 707, 808, 909, 1010	all six	60

Training and evaluation seeds are disjoint. The static snapshot pools are used only for the §3.3 marginals, and only the six pools that `data_manifest.json` labels `train`; `public_07/public_08` (validation) and `public_09/public_10` (development test) are untouched. Rows are never split by pair or person across train and test — the split is by *world*, which is the only unit that respects the dependence structure, since all 200 members of a pool share one generator draw.

Every method sees the identical (seed, variant) worlds. Policy memory is empty at every episode start, matching the assessment protocol.

Randomness is fully declared. The world generator is seeded by the simulator from (seed, variant) and we never touch it. Inside the policy, every random draw uses a fixed generator seeded only from public inputs: the shuffled feasible matcher uses `random.Random(17 + day)` and the residual-soft ask tiebreak uses `random.Random(917 + seed)`. There is no wall-clock, no `os.urandom` and no unseeded random call anywhere in the decision path, so a rerun is bit-identical — which is what makes the `defaulttask_core4_maxw/fullbudget_unranked_maxw` determinism check in §6.2 meaningful. Statistical resampling uses `random.Random(20261009)` for bootstrap intervals, fixed independently of the policy. Nothing in the policy reads a world seed to infer an outcome; the seed enters only as an offset into our own shuffle, and the same offsets are used for every configuration so that ties in §6 are ties of rule rather than of draw. All results are in-process against the public simulator; these are **not** official subprocess or Docker timings, and §6.7 reports the wall-clock cost so the reader can judge the 10-second limit separately.

5.2 Hypotheses and falsification criteria

Each hypothesis is stated with the test that would kill it. We commit to these before reporting the head-to-head numbers.

#	Hypothesis	Test	Falsified if
H1	Targeting clarification at expected matching-unlock value beats count-based potential asks and beats unranked full-budget asks, at a fixed matcher.	Paired seed-block contrast, same worlds	The 95% seed-block CI of the difference includes ≤ 0 at ≥ 10 seeds per family
H2	Any asking policy beats no clarification on coverage and MSMI.	Paired contrast against <code>no_ask</code>	CI includes 0
H3	Adapting on dense directional responses is no worse than a static prior anywhere, and better in drift and shift.	Paired contrast overall and per family	Worse overall with a CI excluding 0, or no gain in drift/shift
H4	In supply-rich families, clearing every day beats clearing every k days; in sparse it does not.	Batching ablation, $k \in \{2, 3, 5\}$	The ordering is the same in both regimes
H5	Residual soft-field clarification does not improve MSMI in development, because at median degree 2 the matching is close to forced and better weights have little room to act.	Paired contrast, <code>unlocksoft</code> vs <code>unlock</code> at fixed matcher	CI excludes 0 in either direction — a real effect, either way, falsifies the near-forced reading
H6	Coverage of any valid policy converges to the §3.1 ceiling; the remaining gap is under 2 percentage points in development.	Coverage ÷ ceiling per family	A policy exceeds the computed ceiling, which would mean the ceiling is wrong

H5 and H6 are stated as *negative* predictions on purpose. A research note whose hypotheses can only be confirmed is not doing much, and the structural analysis in §3 makes a falsifiable claim about what cannot be improved.

5.3 Baselines and ablations

The three required baselines run through the same harness on the same worlds: supplied **greedy** (default asks + starter greedy), **no-clarification**, and **random-feasible**. The incumbent from our own pilot, **potential-ask + 7-soft-field max-weight**, is included as a fourth reference because it is the configuration this note has to beat or explain.

Ablations are factorial in structure, one component changed at a time:

Ablation	Changes	Isolates
<code>unlock_greedy</code> vs <code>greedy</code>	ask rule only	the value of unlock-value targeting

Ablation	Changes	Isolates
defaulttask_core4_maxw vs greedy	weight + allocation only	the value of signed core-4 weighting and global matching
fullbudget_unranked_maxw vs unlock_prior_maxw	ranking only, budget fixed	whether <i>which</i> members to ask matters once the budget is spent
defaultsoft_prior_maxw vs unlocksoft_prior_maxw	ranking of soft asks only	whether unlock ranking matters for soft fields too
unlock_prior_online_day vs unlock_prior_maxw	online adaptation	H3
no_ask vs greedy	clarification entirely	H2, and the required no-clarification ablation

5.4 Statistics

Outcomes are counts, so we treat the *seed block* — one seed’s mean across the six families — as the resampling unit, which preserves the within-seed correlation across families that the official score induces by averaging them.

- **Primary score** = unweighted mean of six family means, exactly as specified. Denominators are never pooled across families and no best seed is ever selected.
- **95% CI** = percentile bootstrap over the ten seed blocks, 10,000 draws, analysis seed 20261009.
- **Paired test** = exact two-sided sign-flip permutation over the ten seed-block differences (2^{10} arrangements, so the smallest attainable p is 0.002).
- **W/T/L** counts over all 60 paired episodes, reported alongside, because with ties dominant, a block-level permutation test and a per-episode count can disagree, and both are informative.
- **No multiplicity correction is applied.** We ran a dozen contrasts. Reported p-values are exploratory; a Bonferroni-adjusted threshold at $0.05/12 \approx 0.004$ is attainable in principle by the exact sign-flip test (smallest reachable p ≈ 0.001 at 2^{10} arrangements) but demanding — it requires near-unanimous seed-block agreement — which is itself the honest summary of how much evidence ten public seeds can carry.

5.5 Power

We measured the per-episode SD of MSMI/100 **within** each method rather than pooling across methods, since pooling inflates it with between-method variance. Across all 17 configurations at 60 episodes each the mean within-method SD is **0.457** (range 0.314 for no_ask to 0.586 for lab_potential_core4_maxw); pooling every episode gives 0.467. Our earlier internal estimate of 0.57 was too high, and using it would have understated the power of a given design by about 35%.

With $n \approx 2\sigma^2(z_{0.975} + z_{0.80})^2/\Delta^2$ at $\sigma = 0.457$:

Gap to detect (MSMI/100)	Episodes per arm	Equivalent seeds $\times$ 6 families
0.50	14	2.3
0.30	37	6.2
0.20	83	13.8
0.15	146	24.3
0.10	329	54.8

Four consequences we design around rather than complain about:

1. **Our design is underpowered for its own headline.** Sixty episodes per arm resolve a gap of about **0.234**. Only one contrast among asking policies that we report is that large, and it is the best-of-seventeen cell (§6.2a).
2. **The official assessment is barely better.** Twenty private seeds per family gives 120 episodes, resolving about **0.165** — comparable to the entire spread between the best and worst asking policies we measured (0.400 to 0.592). With a mean of 0.86 MSMI events per episode, the private ranking of any one submission rests on roughly 104 events. Two competent submissions can be separated by noise, and the tie-break ladder (coverage, then mutual acceptances, then lower ask cost, then lower inference time) will decide more of the ranking than the primary score will.
3. **A three-seed pilot resolves about 0.43** — larger than the distance between the best method we tested and the starter greedy baseline. This is why we do not build narrative on the pilot’s ordering, and why the earlier “4-phase hybrid” result (+1 event over its own fixed control, selected as the best of several screened schedules) is reported as exploratory only.
4. Episode scores are not Gaussian; they are scaled near-Poisson counts where one event moves the score by 0.5 and most episodes are 0, 0.5 or 1.0. The formula above is therefore an approximation, and we report exact sign-flip p-values and W/T/L counts alongside every interval rather than relying on the normal approximation alone.

5.6 Validity threats we can name

Threat	Effect	Mitigation
Public seeds only	Private worlds differ; family-level behaviour may not transfer	Report per family, never only the primary score

Threat	Effect	Mitigation
Winner's curse	Screening many configurations inflates the best one	Report every configuration screened, including the losers
In-process timing	Not the official 10 s / 2-core / 1 GiB container limit	Report wall clock separately; container test is Round 2 work
Unadjusted multiplicity	p-values are optimistic	State the number of contrasts; give W/T/L as well
Rare events	A zero is not an equivalence result	Report counts and CIs, never only means
Label-cutoff bug (found in our own code)	Online learners may have trained on wrong labels	See §7.5

6. Results

Scale. 1,020 head-to-head episodes across 17 configurations, 10 public seeds and 6 families, plus 180 oracle-ladder episodes, 108 logging episodes and 18 equivalence cases. Every episode in every run completed with valid simulator actions: **zero invalid episodes out of 1,326**. All results are in-process against the public simulator on public seeds — *not* official subprocess or Docker timings.

Replication. Before reporting anything new we checked that our independent harness reproduces the previously published ten-seed screen. It does, exactly:

Cell	Published	Ours	Published events	Our events
default asks + starter greedy	0.350	0.350	42	42
no clarification + starter greedy	0.167	0.167	20	20
default asks + random feasible	0.358	0.358	43	43
default asks + 7-soft max-weight	0.408	0.408	49	49
potential asks + 7-soft max-weight	0.525	0.525	63	63

Agreement to the event count on all five cells means the new rows below are comparable to the earlier screen, and that the harness is not introducing its own variance. §6.9 reports a stronger version of the same check against the organisers' reference `policy.py`.

Method	MSMI/100	95% seed-block CI	Events	Coverage	Ceiling	% of ceiling	Assignments	Ask cost
lab_ potential _core4_ maxw	0.592	[0.392, 0.808]	71	0.343	0.352	98%	77.1	197
lab_ potential _maxw	0.525	[0.408, 0.633]	63	0.343	0.352	98%	76.4	197
unlock_ soft7_ maxw	0.525	[0.375, 0.733]	63	0.342	0.352	97%	76.1	195
unlock_ core4_ maxw	0.492	[0.367, 0.650]	59	0.341	0.352	97%	77.2	195
defaulttask _core4_ maxw	0.483	[0.358, 0.600]	58	0.342	0.352	97%	76.9	197
unlock_ greedy maxw	0.475	[0.317, 0.633]	57	0.343	0.352	98%	76.2	195
unlocksoft _prior_ maxw	0.417	[0.233, 0.617]	50	0.341	0.352	97%	77.8	582
defaulttask _soft7_ maxw	0.408	[0.308, 0.500]	49	0.342	0.352	97%	76.6	197
unlock_ prior_ maxw	0.400	[0.250, 0.592]	48	0.340	0.352	97%	77.5	195
unlock_ prior_ online_ day	0.400	[0.250, 0.592]	48	0.340	0.352	97%	77.6	195
random_ feasible	0.358	[0.267, 0.458]	43	0.342	0.352	97%	76.4	197
greedy	0.350	[0.233, 0.467]	42	0.343	0.352	97%	77.3	197
no_ask	0.167	[0.050, 0.292]	20	0.120	0.352	33%	20.9	0

60 episodes per configuration, ten seeds × six families. Ceiling is the §3.1 structural maximum on the same worlds; % of ceiling is coverage divided by it, pooled across families as a summary figure — the per-family ceilings differ (§3.1), and `results/primary_summary.csv` additionally carries the per-seed coverage-to-ceiling ratios, which are the primary utilisation statistic. The four configurations

omitted from this table — `fullbudget_unranked_maxw` (identical to `defaulttask_core4_maxw` by construction), `lab_potential_prior_maxw`, `defaulttask_prior_maxw` and `defaultsoft_prior_maxw` — are in `results/primary_summary.csv`, along with per-method median and 90th-percentile waiting times.

6.1 The headline is a negative result, and it is the useful one

Read down the “Assignments” column. Across the sixteen asking configurations the total moves from 4,555 to 4,681 introductions over 60 episodes — a range of 126 against a total near 4,600. Coverage moves between 0.3399 and 0.3432. And every one of them sits at **97-98% of the structural coverage ceiling** computed in §3.1 — including the unmodified starter greedy baseline.

This confirms H6 and it changes what the problem is. Clarification is not a lever that skilled targeting pulls further than unskilled targeting; it is a **step function**. Asking at all takes coverage from 0.120 to ~0.342 and MSMI/100 from 0.167 to ~0.45. Asking *cleverly* takes it nowhere measurable, because there is nowhere left to go: the ceiling is a property of the generated world, not of the policy.

Only `no_ask` sits away from the ceiling, at 33% of it. Its contrast against the starter greedy baseline is **+0.183 MSMI/100, 95% seed-block CI [+0.042, +0.325], sign-flip p = 0.055, 22 wins / 33 ties / 5 losses**, 42 events against 20. H2 is supported (the bootstrap CI excludes zero; the exact sign-flip p is 0.055), and it is supported robustly: we enumerated all 136 pairwise contrasts among the 17 configurations, and **all sixteen comparisons between an asking policy and `no_ask` have an interval excluding zero**, from +0.183 for the unmodified starter greedy baseline to +0.425 for our best cell.

The same enumeration is what makes the negative result precise. Among the thirteen configurations that both ask *and* match, only three of the 78 pairwise intervals exclude zero, and all three are the same comparison in different cells: a **fitted-prior edge weight loses to a hand-coded one** (`lab_potential_core4_maxw` – `lab_potential_prior_maxw` +0.125, CI [+0.067, +0.192]; `lab_potential_core4_maxw` – `defaulttask_prior_maxw` +0.192, CI [+0.025, +0.367]; `lab_potential_core4_maxw` – `defaultsoft_prior_maxw` +0.183, CI [+0.025, +0.350]). Every other contrast between two competent asking policies straddles zero. So the study can reliably separate *asking from not asking*, and *weight coding from fitted probability*, and it cannot reliably separate one sensible asking policy from another — which is exactly what §5.5’s power analysis predicts and exactly what the thesis claims. The full enumeration (136 contrasts, 29 excluding zero, each with its seed-block bootstrap interval and sign-flip p) is persisted in `results/analysis.json` under `all_pair_contrasts`, so this paragraph is checkable rather than asserted.

Everything else is pair quality on a fixed volume, and pair quality is bounded twice over: by a mean-degree-2.5 matching with little room to choose, and by a funnel in which the two largest losses — non-response and the on-time double-Yes — are outside the policy’s control.

6.2 The factorial

Allocation is general-graph maximum-weight matching in every cell except the two greedy ablations noted below. Cells show MSMI/100 with total events over 60 episodes in brackets.

Ask rule	7-soft agreement count	Signed core-4	Fitted logistic prior
Default / unranked	0.408 (49 ev, n=60)	0.483 (58 ev, n=60)	0.400 (48 ev, n=60)
Count-based potential (incumbent)	0.525 (63 ev, n=60)	0.592 (71 ev, n=60)	0.467 (56 ev, n=60)
Unlock-value (ours)	0.525 (63 ev, n=60)	0.492 (59 ev, n=60)	0.400 (48 ev, n=60)

Allocation is general-graph maximum-weight matching (`networkx.max_weight_matching`) in every cell. Two allocation ablations sit outside the grid: starter greedy ordering with default asks scores 0.350 (42 events) and with unlock asks 0.475 (57 events).

Paired contrasts inside the grid (same worlds, seed-block bootstrap, exact sign-flip):

Contrast	Mean diff 95% CI	p	W/T/L	Events
Global matching vs greedy ordering (soft-7 weights, default asks)	+0.058 [-0.058, +0.167]	0.445	14/40/6	49 vs 42
Signed core-4 vs 7-soft agreement count (default asks)	+0.075 [-0.025, +0.192]	0.328	14/39/7	58 vs 49
Fitted prior vs signed core-4 (default asks)	-0.083 [-0.200, +0.042]	0.266	7/39/14	48 vs 58
Count-based potential asks vs default asks (soft-7)	+0.117 [-0.017, +0.233]	0.160	22/28/10	63 vs 49

Contrast	Mean diff	95% CI	p	W/T/L	Events
Unlock-value asks vs count-based potential asks (soft-7)	-0.000	[-0.167, +0.200]	1.000	17/26/17	63 vs 63
Unlock-value asks vs default asks (signed core-4)	+0.008	[-0.183, +0.192]	1.000	20/25/15	59 vs 58
Unlock-value asks vs default asks (fitted prior)	-0.000	[-0.142, +0.150]	1.000	15/32/13	48 vs 48
Coding + matching, default asks (vs official greedy)	+0.133	[+0.008, +0.258]	0.102	20/33/7	58 vs 42
Coding + matching, unlock asks (vs official greedy)	+0.142	[-0.067, +0.358]	0.281	23/23/14	59 vs 42
Pre-specified: incumbent vs official greedy	+0.175	[+0.092, +0.258]	0.016	23/29/8	63 vs 42
Value of clarification (required ablation)	+0.183	[+0.042, +0.325]	0.055	22/33/5	42 vs 20

Three readings matter more than any single cell.

(a) The best cell is potential asks + signed core-4, at 0.592 (71 events). Against the official greedy baseline that is **+0.242 MSMI/100, CI [+0.125, +0.383], p = 0.008, 27 wins / 27 ties / 6 losses**; against random-feasible +0.233, CI [+0.050, +0.458], p = 0.064; against no-clarification +0.425, CI [+0.242, +0.625], p = 0.002. It is also above the incumbent 0.525.

We state the selection problem before anyone else can: this cell is the **best of seventeen screened configurations**, so its own p-value is not the relevant one. Bonferroni across the twelve contrasts pre-specified in the table above puts the adjusted threshold at $0.05/12 \approx 0.004$, and across all 136 pairwise contrasts at $0.05/136 \approx 0.0004$; p = 0.008 clears neither. With 60 episodes per arm the design resolves about 0.234 (\$5.5) and the effect is 0.242 — right at the resolution limit. The defensible claim is: *a configuration combining count-based potential asking with signed core-4 weighting and global matching scored 0.592 on ten public seeds, nominally above both the official greedy baseline and our own incumbent, and this must be re-tested on fresh seeds before it is believed.* It is a candidate for Round 2, not a result.

(b) The matcher improvement decomposes into two halves, neither significant alone. Changing only the allocation, from the starter's greedy ordering to global maximum-weight matching at fixed 7-soft weights and fixed default asks, gives **+0.058, CI [-0.058, +0.167], p = 0.445, 14/40/6**. Changing only the weight coding, from a 7-field agreement count to signed core-4 scoring, gives **+0.075, CI [-0.025, +0.192], p = 0.328, 14/39/7**. Together the two give +0.133, CI [+0.008, +0.258], p = 0.102, 20/33/7 — measured directly as `defaultask_core4_maxw - greedy`, and equal to the sum of the two halves to three decimal places. The two effects are roughly additive and each is about half the combined size, so neither can be credited on its own. The coding change is not a model: it corrects two implementation errors — treating a *known disagreement* as equivalent to *no information*, and allowing weights to go non-positive so the matcher discards feasible edges whose true outcome probability is small but strictly positive.

(c) Ask targeting does not help, and the fitted prior actively hurts. At a fixed matcher, unlock-value ranking against count-based potential asking gives -0.000 (CI [-0.167, +0.200], 17/26/17, both at 63 events); against default asking it gives +0.117 at 7-soft weights but +0.008 at core-4 and -0.000 at the fitted prior. **H1 is not supported.** Worse, the fitted logistic prior is *below* the hand-coded signed rule in all three ask cells, and in the potential-ask cell the gap has an interval excluding zero: **-0.125, CI [-0.192, -0.067], p = 0.008, 2 wins / 43 ties / 15 losses**. Given twelve pre-specified contrasts this too is not conclusive, but the direction is consistent across all three ask rules, which is harder to dismiss than a single cell.

We think we understand why, and the explanation is more interesting than the number. A calibrated probability is the *theoretically correct* edge weight for maximising expected MSMI on a single day, since the objective is linear in the chosen edges. But maximum-weight matching with `maxcardinality=False` is **scale-sensitive**: it trades “match more pairs” against “match better pairs”, and that trade depends on the spread of the weights, not only on their order. Our prior maps every feasible pair into $\hat{p} \in [0.23, 0.56]$ — a band so narrow that the optimiser behaves almost like maximum-cardinality matching and largely ignores quality. The signed core-4 rule spans [0.06, 0.98] and the

7-soft count spans $[0.01, 7.01]$, so both impose a real quality/cardinality trade. That a correctly calibrated probability underperforms a mis-scaled heuristic is **consistent with a weight-scale mechanism** — and, if the scale story is not the whole of it, with the stronger claim that the myopic single-day linear objective is itself misspecified. The two explanations are separable, and we say so plainly: re-running the fitted prior under a monotone spread (or rank) transform that preserves its ordering would isolate the scale effect; if the prior then recovers, the misspecification reading weakens. That discriminating test is planned for Round 2 (§8.1). The sequential structure (eight-day busy windows, non-repeatable pairs, permanent retirement on success) means an edge’s value is not its one-day probability; that is the concrete argument for the look-ahead work in §8.1, and it is a better reason to keep going than any score in this table.

6.3 Scenario-level results

Method	development	sparse	cold start	delayed	shift	drift	Primary
lab_potential_core4_maxw	0.850	0.150	0.350	0.800	0.650	0.750	0.592
lab_potential_maxw	0.800	0.100	0.400	0.550	0.550	0.750	0.525
unlock_soft7_maxw	0.700	0.050	0.350	0.850	0.500	0.700	0.525
unlock_core4_maxw	0.650	0.100	0.300	0.650	0.650	0.600	0.492
defaulttask_core4_maxw	0.550	0.050	0.600	0.650	0.450	0.600	0.483
unlock_greedy	0.600	0.050	0.450	0.550	0.600	0.600	0.475
unlocksoft_prior_maxw	0.450	0.050	0.350	0.700	0.500	0.450	0.417
defaulttask_soft7_maxw	0.450	0.000	0.600	0.600	0.400	0.400	0.408
unlock_prior_maxw	0.500	0.050	0.350	0.550	0.500	0.450	0.400
unlock_prior_online_day	0.500	0.050	0.350	0.550	0.500	0.450	0.400
random_feasible	0.500	0.050	0.350	0.400	0.350	0.500	0.358
greedy	0.350	0.050	0.550	0.400	0.400	0.350	0.350
no_ask	0.300	0.000	0.000	0.150	0.250	0.300	0.167
coverage	0.399	0.177	0.341	0.399	0.399	0.399	
ceiling (§3.1)							

- **Sparse is 0.00-0.15 for every configuration.** With 23.3 usable pairs per world and the measured funnel, the expected value is ~ 0.12 (§7.1). The published zeros and our near-zeros are both consistent with a supply-limited family sampled ten times; a zero is not an equivalence result, and neither is 0.15.
- **Cold start still favours untargeted asking, and by a wide margin.** defaulttask_core4_maxw and fullbudget_unranked_maxw reach 0.600 where unlock_core4_maxw reaches 0.300 and unlock_prior_maxw 0.350 — a full reversal of the development ordering (about 12 vs 6 events in this one family; exploratory). §7.3 gives a working explanation.
- **shift and drift do not reward the offline prior**, exactly as §4.5 predicted: the prior pools stationary and non-stationary families, so its coefficients are a compromise that fits neither well. The online correction was supposed to repair this and did not (§6.5).

6.4 The funnel

Method	Assigned	Both responded	Mutual accept	Date	Date $\leq 30d$	Both on time	MSMI
lab_potential_core4_maxw	4624	2753	704	571	567	174	71
lab_potential_maxw	4584	2717	727	560	554	156	63
unlock_soft7_maxw	4566	2687	676	549	543	145	63
unlock_core4_maxw	4629	2819	714	562	560	167	59
defaulttask_core4_maxw	4614	2698	688	534	529	138	58
unlock_greedy	4575	2691	681	544	539	133	57

Method	Assigned	Both responded	Mutual accept	Date	Date $\leq 30d$	Both on time	MSMI
unlocksoft	4666	2747	676	533	532	140	50
_prior_maxw							
defaulttask	4594	2636	669	515	511	122	49
_soft7_maxw							
unlock	4652	2869	722	573	572	144	48
_prior_maxw							
unlock	4658	2859	715	577	576	143	48
_prior_online_day							
random	4583	2717	672	535	530	114	43
feasible							
greedy	4641	2710	645	511	508	99	42
no_ask	1253	750	180	152	152	47	20

Method	P(both respond)	P(mutual responded)	P(date mutual)	P(on-time both date)	P(MSMI assigned)
lab_potential_core4_maxw	0.595	0.256	0.811	0.305	0.0154
lab_potential_maxw	0.593	0.268	0.770	0.279	0.0137
unlock_core4_maxw	0.609	0.253	0.787	0.297	0.0127
defaulttask_core4_maxw	0.585	0.255	0.776	0.258	0.0126
unlock_greedy	0.588	0.253	0.799	0.244	0.0125
unlocksoft_prior_maxw	0.589	0.246	0.788	0.263	0.0107
unlock_prior_maxw	0.617	0.252	0.794	0.251	0.0103
random_feasible	0.593	0.247	0.796	0.213	0.0094
greedy	0.584	0.238	0.792	0.194	0.0090
no_ask	0.599	0.240	0.844	0.309	0.0160

Volume is identical; only conversion differs, and the conversion differences live in the noisiest stage. P(both respond) is 0.585–0.617 across methods and P(date | mutual accept) is 0.75–0.80 everywhere, matching the **0.769** we measured on the 1,162 mutual acceptances in our randomised logging set — both are constants of the world, not policy variables. P(mutual accept | both responded) spans 0.246–0.268, a 2.2-point absolute band. P(both on time | date) spans 0.228–0.305 across the thirteen asking-and-matching configurations, on 515–577 dates; the binomial SD of a single such rate at $p = 0.26$, $n \approx 546$ is 0.019, and the expected range of thirteen draws with that SD is about 0.06 — close to the 0.077 observed. The spread across methods at the final stage is therefore of the same order as sampling noise, though we do not claim it is entirely noise.

The consequence is uncomfortable and we state it plainly: **the stage that separates the top configuration from the bottom is the one least connected to anything a matcher controls.** We therefore do not interpret the 71-vs-42 event gap as proof that any weighting scheme is better. We interpret it as a candidate that has survived one screen and needs a second.

The loss structure itself is unambiguous. Averaged across the sixteen asking configurations, 60 episodes produce **4,613** assignments, of which **2,733** get two responses (59.2%), **690** reach mutual acceptance, **546** produce a date (542 inside the 30-day window), **138** have both members answer the second-meeting question on time, and **54** qualify as MSMI. Of the **4,559** assignments that do not become an MSMI, **1,880** (41%) are lost at the response stage, **2,043** (45%) at mutual acceptance, **144** (3%) at the date stage, **408** (9%) at the on-time answer stage and **84** (2%) at the final both-Yes step. The response and on-time stages are outside the policy's control; the on-time stage is capped at $0.6^2 = 0.36$ by the 1–5 day answer delay against the 3-day window no matter what the policy does. The mutual-acceptance stage is where pair selection acts, and §6.6 shows that even perfect information raises its rate only about 12% in relative terms.

6.5 Online adaptation changed nothing

unlock_prior_online_day and unlock_prior_maxw produced **identical MSMI counts in all 60 episodes: 0 wins, 60 ties, 0 losses, mean difference exactly 0.000.** This is not a null caused by the adaptation failing to run. Instrumented on a drift world, the online learner harvested 118 directional observations, performed 22 ridge-anchored refits, moved the intercept from -1.006 to $+0.203$ and learned a day coefficient of -0.319 — the right sign for drift, whose measured acceptance rate falls from 0.263 before day 30 to 0.164 from day 45.

It changed nothing because **the matching is invariant to it.** Refitting moves the intercept (irrelevant to a ranking), rescales coefficients roughly proportionally (nearly irrelevant), and only occasionally reorders two edges whose weights are close — and at median degree 2, reordering rarely

changes which matching is optimal. H3’s “no worse anywhere” holds trivially; “better in drift and shift” is **falsified as implemented**.

The honest reading is that online adaptation of *edge weights* is the wrong place to spend adaptivity here. If a policy is to react to drift it has to change a decision that weights do not control — how many pairs to propose, whether to wait, or which members to clarify. We also record a defect we found while instrumenting this: the offline prior targets $P(\text{mutual acceptance} \mid \text{both responded})$, base rate 0.254, while the online update fits $P(\text{directional Yes} \mid \text{responded})$, base rate ~ 0.55 . Two different targets, which is why the intercept moves so far. It does not affect ranking, but a Round 2 implementation must fit one target consistently.

6.6 Oracle ladder (analysis-only)

The R4 and R5 rungs write disclosed answers directly into the simulator’s member records. No policy may do this and none of ours does. Per our own audit rule the latent-pair-probability rung was dropped, because the simulator does not expose latent pair probabilities and reconstructing its outcome function would not be a defensible reference.

Rung	cold_start	development	sparse	Mean	Assignments	Coverage
R0 no clarification + greedy	0.000	0.300	0.000	0.100	14.6	0.091
R1 default asks + random feasible	0.350	0.500	0.050	0.300	61.9	0.297
R2 official greedy baseline	0.550	0.350	0.050	0.317	62.5	0.298
R3 our policy with the fitted prior (unlock asks + prior weights + global matching)	0.250	0.450	0.050	0.250	63.2	0.297
R4 oracle: free complete clarification (hard and soft)	0.100	0.500	0.100	0.233	62.9	0.298
R5 oracle: R4 + maximum-cardinality allocation	0.150	0.450	0.100	0.233	63.1	0.298

180 oracle-ladder episodes over 10 seeds. R4/R5 are **analysis-only**: they write disclosed answers straight into the simulator’s member records, which no policy may do and none of ours does. The latent-pair-probability rung was dropped (§4.7).

This is the most important table in the note. **From R1 (uniform random feasible pairs) to R5 (an oracle given free, complete, instantaneous disclosure of every non-declined hard and soft field, plus maximum-cardinality allocation), coverage moves from 0.297 to 0.298 and assignments from 61.9 to 63.1.** The entire ladder spans 14–19 MSMI events over 30 episodes per rung, where the Poisson SD at 16 events is 4. Only R0 — no clarification at all — is separable, at 6 events.

Perfect information does buy something measurable at the intermediate stage: $P(\text{mutual accept} \mid \text{both responded})$ rises monotonically from 0.245 at R1 to 0.272 at R4 and 0.275 at R5, about a 12% relative improvement on $\sim 1,100$ both-responded pairs per rung. **It does not convert.** MSMI per assignment is 0.0097 at R1, 0.0074 at R4 and 0.0074 at R5, because the downstream stages move the other way within their own noise.

Two caveats on this table, because it is the one most easily misread. First, R3 is *not* our best configuration. The ladder was built before the factorial completed, so R3 carries the fitted logistic prior as its edge weight — the weight rule §6.2c shows is the weakest of the three, and the one that loses in cold start (§6.3). R3 scoring 0.250 against R2’s 0.317 is therefore a reproduction of the §6.2c finding inside the ladder, not evidence against the method; the configuration that scores 0.592 in §6.2 was never run on these three families, and we do not claim it would beat R2 here. Second, the ladder covers only development, sparse and cold_start, so its means are **not comparable** to the six-family primary scores in §6.2 — it is an internal comparison across rungs on identical worlds, and nothing else. What survives both caveats is the R1-to-R5 comparison, which involves no weighting choice we would defend and no family we excluded: uniform random feasible pairs versus an all-knowing oracle move coverage by 0.001 and assignments by 1.2.

Two conclusions follow, and they are the ones we would want a reviewer to take away:

1. **Information is not the binding constraint.** No clarification strategy, however well targeted, and no pair-outcome model, however well calibrated, can close a gap that an oracle with perfect answers does not close. This is why the targeting results in §6.2 are null and why we should have expected them to be.
2. **A surrogate can move while the primary does not.** Mutual acceptance — the first official tie-break — improves 12% from R1 to R5 while MSMI is flat. Optimising or reporting the surrogate without the primary would have produced a confident false claim. The spec’s insistence that assignment, acceptance, date and second-meeting intention be reported separately is doing real work here.

6.7 Cost, waiting times and coverage accounting

Metric	Hard-only ask rules (n=14)	Residual-soft ask rules (n=2)	no_ask
Ask cost per episode (ceiling 720)	196	598	0
Mean first-introduction wait (days)	6.32	6.15	3.95
Median first-introduction wait, of served (days)	3.97	3.64	0.72
90th-pct first-introduction wait, of served (days)	16.02	15.90	13.77
Unserved members of 200	131.6	131.8	176.0
Missing feedback events per episode	39.5	39.7	10.2
Coverage	0.342	0.341	0.120
In-process seconds per episode	6.74	5.83	6.44

Waiting times are computed over **served** members only, as the definition of a first-introduction wait requires; unserved members are reported separately two rows below and stay in every coverage denominator. Each cell is the mean over that group’s 60 episodes of a per-episode statistic. Per-method values are in results/primary_summary.csv.

Ask spend is ~195–197 units per episode for every hard-only rule and ~582–614 for rules that convert the residual into soft-field answers — against a 720-unit ceiling. **The budget is not binding and never was.** With 31.9% of development members permanently decline-blocked and 36.4% already complete at arrival, the askable population runs out long before the units do; §6.8 shows what happens when the residual is spent anyway. Any narrative about “spending the clarification budget wisely” should be replaced by one about whether the residual can buy anything at all.

Waiting times are reported with the unserved count, as the spec requires, and members who can never be served stay in the denominator. no_ask has the shortest mean wait (3.95 days) precisely because it serves so few: it introduces the members who happen to be pre-disclosed and leaves about 176 of 200 unserved. A low waiting time is not a good outcome when it is bought with a 12% coverage.

6.8 Where the residual budget goes

Contrast	Mean diff	95% CI	p	W/T/L	Events	Ask cost A vs B
Residual soft asks vs hard-only (unlock ranking fixed)	+0.017	[-0.083, +0.158]	0.961	9/41/10	50 vs 48	582 vs 195
Unlock-ranked soft asks vs member-order soft asks	+0.008	[-0.200, +0.208]	1.000	17/28/15	50 vs 49	582 vs 614
Online dense-proxy adaptation vs static prior	+0.000	[+0.000, +0.000]	1.000	0/60/0	48 vs 48	195 vs 195

Spending the residual ~390 units on soft-field answers raises realised ask cost from 195 to 582 and changes MSMI/100 by **+0.017, CI [-0.083, +0.158], p = 0.961, 9 wins / 41 ties / 10 losses** — two events across 60 episodes. Ranking those soft asks by unlock value rather than taking them in member order changes nothing (+0.008, p = 1.000, 17/28/15).

H5 is not rejected (the interval is consistent with the near-forced reading but too wide to establish equivalence), and the oracle ladder in §6.6 explains it: soft answers improve w_{ij} , and R4 shows that even *perfect* soft answers improve the mutual-acceptance rate by only ~12% and the primary score not at all. At median degree 2 the maximum-weight matching is close to forced, so knowing the weights more precisely rarely changes which pairs are selected. The information is real and the

decision it would inform does not exist.

We regard this as the most transferable result in the note for anyone building on this kit: **before designing a clarification policy, measure the degree distribution of the graph it is meant to improve.** If the matching is forced, information has no leverage, and the budget should be spent elsewhere or not at all.

Where the ask budget actually goes. Every configuration spends about 195 of its 720 budget units (\$6.7), which reads like under-spending until you measure the askable population directly. The probe below runs each ask rule and records, at eight days, how many of the 200 members have a complete and never-declined hard bundle and how many feasible edges are live between members who are still free.

Members with a complete, never-declined hard bundle (of 200), and live feasible edges between still-free members, averaged over 5 seeds:

Family	Ask rule	d0	d5	d10	d15	d20	d30	d45	d59
development	no_ask · members	50	58	63	68	75	75	75	75
development	default_asks · members	54	82	107	128	140	140	140	140
development	fullbudget_unranked · members	54	82	107	128	140	140	140	140
development	potential_ask_count · members	54	82	107	128	140	140	140	140
development	unlock_value · members	54	82	107	127	139	140	140	140
development	unlock_value_plus_soft · members	54	82	107	127	139	140	140	140
sparse	no_ask · members	49	54	61	67	74	74	74	74
sparse	default_asks · members	53	78	105	121	132	132	132	132
sparse	fullbudget_unranked · members	53	78	105	121	132	132	132	132
sparse	potential_ask_count · members	53	78	105	120	132	132	132	132
sparse	unlock_value · members	53	78	105	119	130	130	130	130
sparse	unlock_value_plus_soft · members	53	78	105	119	130	130	130	130
development	no_ask · live edges	11	1	0	1	0	0	2	0
development	unlock_value · live edges	15	4	4	2	5	1	1	2

All five ranked ask rules are within one member of each other at every day after day 10 in both families; live-edge counts for the other ranked rules are in results/ask_timing.json.

Three things follow. **(i)** Asking is *saturated, not rationed*: every ranked rule reaches the same 140-member plateau, and reaches it by day 15–20, leaving 40 days of budget with nobody left to ask. The 60 members who never complete a bundle are the decline-blocked and the never-arrived-observable, and no ordering of asks recovers them. **(ii)** no_ask plateaus at 75, so clarification does move the usable graph from 75 to 140 members — which is exactly the H2 effect and exactly why it is a step function rather than a gradient. **(iii)** Live feasible edges collapse to single digits by day 10 under every rule, including the best ones. The binding constraint is not the ask budget, not the ranking of asks, and not the matcher; it is that the feasible graph is consumed faster than it is replenished. This is the mechanism behind H6 and it is the reason §8.1 prioritises *waiting and batching* — the only lever that acts on the timing of consumption rather than on the quality of a decision made at a moment when there is almost nothing to decide between.

6.9 Reproducibility of the supplied baselines

Because our harness runs in-process rather than through the official subprocess protocol, we verified it against the organisers' reference policy.py decision-for-decision: same world, same days, comparing every ask list and every pair batch, then the final metrics.

Reference mode (policy.py)	Our in-process method	Identical ask lists, pair Cases batches and final metrics
greedy	greedy	6 all identical
no_asks	no_ask	6 all identical
random	random_feasible	6 all identical

Cases cover seeds 101–606 across all six families, 60 days each, comparing every ask list and every pair batch day by day. Overall: **all baselines reproduce the reference exactly**.

7. Failure analysis

The spec requires explaining failures involving sparse supply, withheld answers, delayed feedback and competition for the same candidate, and requires that members with no feasible candidate stay in the coverage denominator rather than being quietly dropped. We take that further: for each failure mode we say what we measured, what we can fix, and what is irreducible.

7.1 Sparse geography — supply, not policy

Every method we and others have run scores near zero in sparse. §3.1 explains why, and the explanation is not a policy defect. Twelve opaque zones with single-zone acceptance put **67.0%** of members at degree zero and leave **23.3** usable pairs in a 200-member pool. At the measured funnel rates (0.5932 both respond, 0.2538 mutual, 0.7685 date, 0.0907 MSMI|date), consuming *every one* of those pairs yields 0.24 expected MSMI events per episode — **0.122 per 100 arrived members**, against **0.619** for a development world at full utilisation. No policy can score meaningfully in sparse; the family is a floor on the primary score, not a test of skill.

Sparse is one sixth of the primary score, so it caps the attainable total. Combining the edge-supply figures with the measured funnel gives an arithmetic bound on the primary score at different levels of pair-selection quality (results/attainable_ceiling.json):

Pair-selection quality	P(mutual both resp.)	MSMI / assignment	dev	sparse	cold	Primary bound
Randomised- logging quality (measured)	0.2538	0.0105	0.619	0.122	0.475	0.512
Incumbent published quality	0.300	0.0124	0.732	0.145	0.562	0.606
Strong selection: the three relevant fields agree	0.400	0.0165	0.976	0.193	0.749	0.808
Best-case selection	0.500	0.0207	1.220	0.241	0.937	1.009

This is a **bound, not a forecast**: it assumes every one of the 118 / 90.6 / 23.3 effective edges is consumed, ignoring arrival and exit timing, the eight-day busy window, permanent retirement after a mutual second-meeting Yes, and the fact that a mean-degree-2.5 matching cannot always take its preferred pairs. Our randomised logging policy in fact consumed 71% of the edge supply in development, shift and drift, 68% in delayed, 79% in cold start and **87% in sparse**, and scored 0.375 overall against the 0.512 bound. So the honest reading is that the *achievable* primary score sits somewhere between roughly 0.5 and 0.8, and that the published incumbent at 0.525 — and our

best screened cell at 0.592 — are both already inside that band. Any claim of a large multiple over the incumbent should be treated with suspicion until the bound is beaten, not merely approached. This is the single most useful number in this note for anyone tempted to over-claim.

What is *not* irreducible: our logging policy consumes ~20 of the 23 available sparse pairs, so the volume lever is nearly exhausted, but pair quality is not — sparse worlds still contain goal- and pace-agreeing pairs and a matcher that finds them may improve the mutual-acceptance rate on the same volume, although the oracle ladder (§6.6) suggests perfect information raises it only about 12% in relative terms. The correct sparse strategy is therefore **selective, not expansive**: with 23 pairs and 200 members, spending clarification to reveal *all* of them is less valuable than revealing the *good* ones early, because members exit and the pool thins. We flag one honest caveat: there are only three distinct sparse worlds among the ten public seeds' generator draws, so per-family sparse estimates carry very little independent information.

7.2 Withheld answers — a permanently excluded third of the pool

A declined hard field is absorbing. **31.9%** of development members and **41.0%** of cold-start members can never appear in a feasible pair. Three consequences:

1. **Coverage must be reported against the ceiling, not against 1.0.** Our measured coverage divided by the §3.1 ceiling is the honest utilisation statistic. Our best configuration reaches coverage **0.465** on development seed 101, which is not “46.5% of members served” but *every one of the 46.5% of that world's members who can ever be served* — the per-seed ceiling there is 0.465. Across the ten development worlds the ceiling ranges 0.370–0.470 and averages 0.399, so a coverage number is uninterpretable without the seed's ceiling beside it.
2. **Clarification budget spent on decline-blocked members is destroyed.** Every ask rule we test filters them out first. This is cheap and no supplied baseline documents how much pool it removes.
3. **Missingness may be informative about availability, not preference.** Since the decline draw is independent of the latent outcome in this generator, we found no evidence that declined members would have been better or worse matches. We verified the analogous question for soft fields and found the opposite of what a naive model assumes: acceptance is *lower* when more soft fields are observed (0.2724 → 0.2317). Any model that encodes observability as a feature will learn a spurious preference. This is a missing-not-at-random *artefact* rather than genuine MNAR structure, and the distinction matters: the fix is to exclude the indicator, not to re-weight by it.

7.3 Cold start — where our own ask rule loses

Cold start is the one family where targeted clarification *underperforms* the untargeted default, and it did so in both our three-seed pilot and the ten-seed scaled run. The following is our working explanation; it is untested, and the shrinkage rule in §8.1 is its test:

- Only **20.6%** of cold-start members have their hard bundle pre-observed, against 36.4% elsewhere, so 79% of the pool is askable and the queue is long.
- **41.0%** are decline-blocked, the highest of any family, so the askable set is both larger and more hazardous.
- Soft-field observation is ~34%, so the weights w_{ij} that drive our ranking $V_i = \frac{1}{3} \sum_j \pi_{ij} \max(0, w_{ij} - u_j)$ are computed from mostly-empty inputs. When w_{ij} is near its intercept for every candidate, the ranking degenerates to $\sum_j \pi_{ij}$ — but with more noise attached, and with the same-zone prefilter interacting badly with a pool where fewer zone answers are known.

This is not a hypothesis we pre-registered — H1 predicted that targeted asking would win everywhere, and cold start is where it fails hardest, in the reverse direction. We report it as the sharpest falsification of H1 in the study: the advantage of a value-ranked ask rule depends on having enough observed soft signal to rank with, and cold start is precisely the family that removes it. The fix is a **shrinkage rule** — as the fraction of observed soft fields falls, blend V_i towards the count-based $\sum_j \pi_{ij}$ and eventually towards uniform ordering. We list this as planned work with a named test (does the cold-start reversal disappear under shrinkage?) rather than claiming it.

7.4 Delayed feedback and the follow-up window

In delayed, the date offset can reach 33 days against the 30-day MSMI window, so some introductions that produce a date and two timely Yes answers still fail the outcome definition. Our label helper now checks `date_day - assigned_day ≤ 30` explicitly and we count the two failure routes separately (`date_in_window` vs `second_on_time` in the funnel table), so a loss is attributed to the right cause.

Right-censoring is handled by construction: labels are harvested only after the 40-day follow-up, and an introduction that has not matured returns None and is excluded from learning rather than coded 0. At the decision horizon, recent introductions are genuinely unresolved and we do not treat them as failures.

The operational consequence is the one that hurts: **feedback arrives too late to help the episode**

that earned it. An introduction made on day 40 has its MSMI label after day 70, past the decision horizon. Only the 7-day directional response is available inside the episode in useful quantity, which is the whole argument for \$4.5.

7.5 An internal audit: the label-cutoff bug

Our earlier lab helper `mature_primary_label` labelled “a date plus two timely second-meeting Yes answers” as an MSMI positive **without checking that the date occurred within 30 days of assignment**. We found one instance in an audited sample (delayed family, seed 303: assigned day 31, date day 62). The simulator’s own scoring was always correct, so no reported *score* was affected; the bug could only corrupt **online learning updates**.

Consequences we accept rather than minimise:

- Every learning-based row from the three-seed pilot (Thompson 0.472, GP-UCB 0.389, GP-mean 0.222) is **provisional** and is not cited as a result here.
- The corrected label is used for all fitting in §4.3 and all results in §6.
- The audit sample was small, so other seeds may have been affected; we did not re-run the old learners, we replaced them.

We report this because the spec asks for honest limitations and because a documented internal audit is more useful to a reviewer than a silently corrected table.

7.6 Competition for the same candidate

With mean degree 2.53 and median degree 2, competition is the norm rather than the exception: a max-weight matching resolves it globally, whereas the starter greedy resolves it by sort order and can consume a member on a mediocre edge before a better one is considered. Two dynamic forms of competition remain unhandled and are named here rather than solved:

- **Temporal competition.** Introducing *i-j* today makes both busy for eight days and consumes the pair permanently. If *i* has a materially better partner who is currently busy or not yet arrived, waiting dominates. We do not yet estimate arrival rates in-episode or compare a waiting rule against daily clearing (H4).
- **Success consumes supply.** A mutual second-meeting Yes retires both members permanently. Maximising MSMI therefore spends the scarcest resource in the system, and in a supply-limited family the optimal policy may be more selective than the single-day linear objective implies. Quantifying this needs a two-stage look-ahead we have not implemented.

7.7 Members who can never be served

They stay in the denominator, as required, and the two exclusion mechanisms must not be added together. Measured across the ten public worlds:

Family	Never introducible	...of which: ≥1 declined hard field	...of which: no feasible partner even under complete disclosure	Overlap
development	60.2%	31.9%	33.9%	5.6 pp
cold start	66.0%	41.0%	32.1%	7.1 pp
sparse	82.3%	32.8%	67.0%	17.5 pp

The two columns are measured on different graphs — the decline count uses the observed questionnaire, the isolation count assumes full disclosure — so they overlap rather than nest, and only their union bounds what is achievable. In sparse the union is 82.3%, which is the whole of \$7.1 in one number.

The unserved count and the first-introduction waiting-time distribution (mean, median and 90th percentile, over served members) are reported by group in §6.7 and per method in `results/primary_summary.csv`, so a reviewer can see the waiting-time cost of selectivity rather than only the success rate.

8. Round 2 plan, disclosure and references

8.1 What is done, what is planned

Item	Status
Structural diagnosis: coverage ceilings, edge supply, binding-constraint histogram	Done (analysis-only)
Empirical hard-constraint pass-probability prior from the six supplied training pools	Done
Self-logged randomised rollouts, 108 episodes, 7,613 labelled introductions	Done
Restricted pair-outcome prior with held-out model selection and a missingness-confound diagnosis	Done
Unlock-value clarification rule with reduced-profit approximation	Done
Residual-budget soft-field clarification rule	Done
Online ridge-anchored adaptation on directional responses, with an in-episode day term	Done and falsified — bit-identical decisions to its own static version in all 60 episodes (§6.5); kept in the harness as an instrumented null

Item	Status
Head-to-head on 10 public seeds × 6 families vs the three required baselines	Done , \$6
Funnel decomposition and per-family scenario reporting	Done , \$6
Oracle ladder up to perfect clarification + maximum-cardinality allocation	Done , \$6.6 (analysis-only)
Cold-start shrinkage rule for the ask ranking (§7.3)	Planned
Waiting/batching ablation, $k \in \{2, 3, 5\}$, and in-episode arrival-rate estimation (H4)	Planned
Exact blossom duals for the reduced-profit term instead of the realised-matching approximation	Planned
Doubly-robust off-policy evaluation on our own randomised logs, with recorded propensities	Planned
Two-stage look-ahead for temporal competition and success-consumes-supply (§7.6)	Planned
JSON stdin/stdout policy adapter, Dockerfile, container-mode timing under 2 cores / 1 GiB / 10 s	Planned
20-seed public runs and a serialised-memory size check against the 1 MiB cap	Planned
Latent-probability oracle rung	Dropped (§4.7)

8.2 Round 2 build order

1. Wrap the winning configuration in the JSON protocol, keeping the starter argument parser, stdout reserved for protocol JSON and diagnostics on stderr. Handle an empty population and an empty feasible graph explicitly.
2. Ship `hard_priors.json` and `outcome_prior.json` as declared inference assets; confirm total image size and serialised memory against the 2 GiB and 1 MiB caps.
3. Containerise and measure. The measured in-process cost per episode-day is small, but the $O(n^2)$ eligibility pass is the term to watch: 200 members means ~20k pair checks per phase, and the 10-second limit includes process startup. The same-zone and age-gap prefilters in §4.2 exist for exactly this reason.
4. Run the cold-start shrinkage rule and the batching ablation; keep whichever survives a paired test at ≥ 20 seeds.
5. Publish per-episode machine-readable results for all four required comparisons on identical seeds and variants, plus the funnel and waiting-time tables, and pin an immutable archive with its SHA-256.

8.3 AI tools disclosure

This note and the accompanying code were produced with the assistance of large-language-model coding agents, used for: reading and summarising the starter kit and its documentation; drafting and debugging the experiment harness, the policy implementations and the analysis scripts; and drafting and critically revising this document. Separate chat-assistant sessions (Claude- and Gemini-family models, via Arena.ai Agent Mode and other tools) additionally contributed ideation, cross-checking of the draft against its own tables, and pre-submission review; every such suggestion was accepted only after arithmetic or artefact cross-check, and the final decisions rest with the team. All experimental numbers reported here were produced by executing the committed Python code against the organisers' public simulator release 1.0.0 on public seeds, and every table is reproducible from the scripts named in §9. No model was used to generate results, and no result was accepted into this note without a corresponding artefact on disk. One model-assisted implementation contained the label-cutoff defect described in §7.5; it was found by human-directed audit of the code against §8 of the problem statement, not by the model. Where an earlier AI-generated draft contained claims contradicted by our own audit — bipartite/Hungarian allocation, zone adjacency, "one introduction per person per day", regret curves, exact value of information, a Thompson-plus-VOI hybrid — those claims are removed here and are not asserted anywhere in this note.

8.4 Synthetic-data disclaimer

All members, profiles, preferences, conversations, introductions and outcomes are newly generated or authored synthetic content. Demographic proportions and outcome behaviour are invented assumptions, not fitted or validated distributions of real people. Simulator performance is an engineering and research test; it is not evidence that any model can predict a relationship, and nothing here should be presented as product effectiveness.

What these invented outcomes cannot establish. Four things specifically. They cannot establish that any feature we weighted — `relationship_goal`, pace, conversation style — predicts real romantic compatibility; the generator decided that, and we recovered the generator's decision, not the world's. They cannot establish that a higher MSMI score is a better product outcome, since MSMI is a synthetic proxy whose own funnel (§6.4) is dominated by invented response and punctuality rates. They cannot establish that clarification is acceptable to real people, since a simulated member never experiences being asked an intimate question, never declines for reasons of safety, and never objects to the timing. And they cannot establish anything about fairness or disparate impact: the generator draws acceptance uniformly and independently of demographics, so our runs contain no bias

to detect, and the absence of a measured disparity here is not evidence of the absence of one in production.

The adapter and validation that real use would require. Concretely, in the order it would have to be built: (1) a schema adapter mapping a live profile store onto the contract's fields, with an explicit, audited rule for what a missing answer means — the three-valued feasibility of §2.2 has to be re-derived against real data-entry behaviour, not assumed; (2) a consent and disclosure layer in front of every clarification ask, with per-question withdrawal, since declined is absorbing here and in production it must also be *respectful*; (3) real eligibility checks against current, verified attributes rather than self-reported ones, with a human review path for anything that excludes a person; (4) outcome labels from actual, consented follow-up rather than a simulated draw, which would take months and would immediately face the censoring problem §2.4 describes — mutual interest removes both people from the pool, so the pairs that succeed stop being observable; (5) prospective validation with a randomised or staggered rollout against a real allocation baseline, powered on the event rate that actually occurs, not on §5.5's synthetic one; and (6) a fairness audit across protected attributes, which requires data this simulator structurally cannot supply. None of that is started. The method in this note is portable — the three-valued feasibility algebra, the supply diagnosis, the matching formulation and the seed-block statistics all transfer — but every number in §6 is a property of a generator we did not write.

8.5 References

All works below were checked against their publishers' records on 9 October 2026; volume, issue and page numbers are as printed there, so the [verify] markers that appeared in earlier drafts of this note are removed. We have not fabricated a citation to fill a gap, and we cite nothing we have not located.

- **Akbarpour, M., Li, S., & Oveis Gharan, S. (2020).** "Thickness and Information in Dynamic Matching Markets." *Journal of Political Economy*, 128(3), 783–815. DOI 10.1086/704761. (Earlier circulated as "Dynamic Matching Market Design.") The paper's central result is that a *Patient* algorithm that waits to thicken the market loses exponentially less than *Greedy* — but **only when the planner can identify agents who are about to depart**; when departure times are not observable, *Greedy* is close to optimal. That precondition is precisely the empirical question our §6.8 probe answers for this simulator: feasible edges here vanish because members are consumed by an introduction, retire permanently after a mutual second-meeting Yes, or age out at day 60, and the first two are visible in the observation contract. H4 (waiting/batching) is therefore not a speculative add-on but the one lever the theory says could still be large, and §8.2 schedules it as the fourth step of the Round 2 build order.
- **Edmonds, J. (1965).** "Paths, Trees, and Flowers." *Canadian Journal of Mathematics*, 17(3), 449–467. DOI 10.4153/CJM-1965-045-4. The blossom algorithm — the correct general-graph maximum-matching primitive for a pool that is not bipartite, and the reason §4.1 uses `networkx.max_weight_matching` rather than a Hungarian assignment.
- **Dudík, M., Langford, J., & Li, L. (2011).** "Doubly Robust Policy Evaluation and Learning." *Proceedings of the 28th International Conference on Machine Learning (ICML '11)*, 1097–1104. The estimator we plan to apply to our own randomised logging rollouts, where the logging propensities are known by construction (§8.1, planned).
- **Russo, D., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018).** "A Tutorial on Thompson Sampling." *Foundations and Trends in Machine Learning*, 11(1). Background for the label-starvation argument in §4.5: with a mean of about 0.9 MSMI events per episode, a posterior over pair outcomes cannot be updated often enough to matter inside a 60-day episode, which is why we adapt on directional responses instead.
- **Hitsch, G. J., Hortaçsu, A., & Ariely, D. (2010).** "Matching and Sorting in Online Dating." *American Economic Review*, 100(1), 130–163. DOI 10.1257/aer.100.1.130. Motivation for treating introductions as a two-sided *allocation* problem with competition for the same candidates (§7.6) rather than as an independent pair-ranking problem.
- **Karp, R. M., Vazirani, U. V., & Vazirani, V. V. (1990).** "An Optimal Algorithm for On-line Bipartite Matching." *Proceedings of STOC '90*, 352–358. DOI 10.1145/100216.100262. Foundational online-matching background; its bipartite one-sided-arrival model and guarantees do not transfer to this simulator's general-graph, reciprocal-feasibility setting, and we do not imply that they do.
- **Gamlath, B., Kapralov, M., Maggiori, A., Svensson, O., & Wajc, D. (2019).** "Online Matching with General Arrivals." *Proceedings of FOCS 2019*, 26–37. DOI 10.1109/FOCS.2019.00011. The closest model-family reference for general arrivals; our daily batch decisions and clarification actions still make the setting distinct.
- **Saar-Tsechansky, M., Melville, P., & Provost, F. (2009).** "Active Feature-Value Acquisition." *Management Science*, 55(4), 664–684. DOI 10.1287/mnsc.1080.0952. Cost-aware information-acquisition context for the clarification problem; an analogy for budgeted questioning, not a direct matching algorithm.

- **Vouchsafe / Romeo & Juliet (2026).** *The One Introduction Problem — final participant specification 1.0.0*, with the bundled docs/DATA_CONTRACT.md, docs/POLICY_INTERFACE.md and docs/SUBMISSION.md. Authoritative for every rule quoted in this note; where an earlier draft of our own work conflicted with it, the specification won and the draft was corrected (§8.3).

8.6 Data and software provenance

- Simulator and data: organisers' public synthetic release 1.0.0; the bundled starter/kit.py is unchanged and records upstream commit a8e26b35118cfa8e886a02f93984923e43ab64f6 (starter/SOURCE.md, starter/LICENSE, starter/DATA_LICENSE.md). Every member and outcome is synthetic (§8.4). No private seeds, private worlds, private organiser files or real-user data were used; the only external data are the six supplied training pools used for the §3.3 marginals.
- Experiment code carries no separate top-level license; the scripts are those named in §9. The three official baselines were additionally re-verified through the organisers' reference evaluator in trusted-local subprocess mode (54/54 episodes identical; research/VERIFICATION_2026_10_09.md).
- The three-seed soft-question screen (72 episodes) and the seven-seed follow-up (42 episodes) were run on the author's Ubuntu VM. The shared bundle contains the aggregate soft-question results but not the VM-generated row-level JSON; this is documented in research/PROVENANCE_AND_ARTIFACT_STATUS.md.

8.7 Computational environment and reproducibility

- All core factorial, head-to-head and oracle-ladder episodes ran in-process; the exact execution host of those runs is not recorded in this bundle and no hardware specification is claimed for them. Wall times in §6.7 are in-process seconds per episode, not official per-invocation or Docker timings. No Docker/container-mode validation has been performed (Round 2 work, §8.2).
- Dependencies are pinned in requirements.txt; the starter kit is standard library only. The working environment for the §9 scripts is recorded in §9 (Python 3.13 on the author's machine); no VM hardware details are claimed.
- Reproducibility anchors: every policy random draw is seeded from public inputs only (§5.1); bootstrap intervals use analysis seed 20261009; reruns are bit-identical by construction.

9. Reproduction

scripts/diagnose_structure.py	# §3.1, §3.2 binding constraints (analysis-only)
scripts/diagnose_supply.py	# §3.1 coverage ceilings and edge supply (analysis-only)
scripts/estimate_priors.py	# §3.3 hard-constraint marginals from training pools
scripts/log_rollouts.py	# §4.3 randomised logging rollouts, training seeds
scripts/fit_outcome_model.py	# staged fits, independence check, calibration
scripts/fit_variant_models.py	# per-family coefficients, missingness confound
scripts/fit_prior.py	# §4.3 specification search and shipped asset
scripts/harness.py	# runner, labels, bootstrap CI, sign-flip test
scripts/policies.py	# all candidate policies
scripts/compare.py	# §6 head-to-head, sharded
scripts/oracle.py	# §6.6 oracle ladder (analysis-only)
scripts/ask_timing.py	# §6.8 askable-population saturation probe
scripts/attainable_ceiling.py	# §7.1 arithmetic bound on the primary score
scripts/test_baseline_equivalence.py	# §6.9 action-for-action check vs starter/policy.py
scripts/analyse.py	# every table in §6 -> results/analysis.json + CSVs
build_note.py	# substitutes results into RESEARCH_NOTE_FINAL.md
make_pdf.py	# Markdown -> RESEARCH_NOTE.pdf (WeasyPrint)

Outputs land in results/: structure_diagnosis.json, supply_bounds.json, logging_rollouts.json, shards, outcome_model.json, variant_models.json, prior_specification.json, main_shard*.json, oracle_shard*.json, analysis.json, and the CSV/Markdown exports. Inference assets are assets/hard_priors.json and assets/outcome_prior.json.

Dependencies: Python 3.13, numpy, networkx, scikit-learn (offline fitting only — no model artefact requires it at inference time), plus markdown and weasyprint to render this PDF. The starter kit itself is standard-library only and is used unmodified.